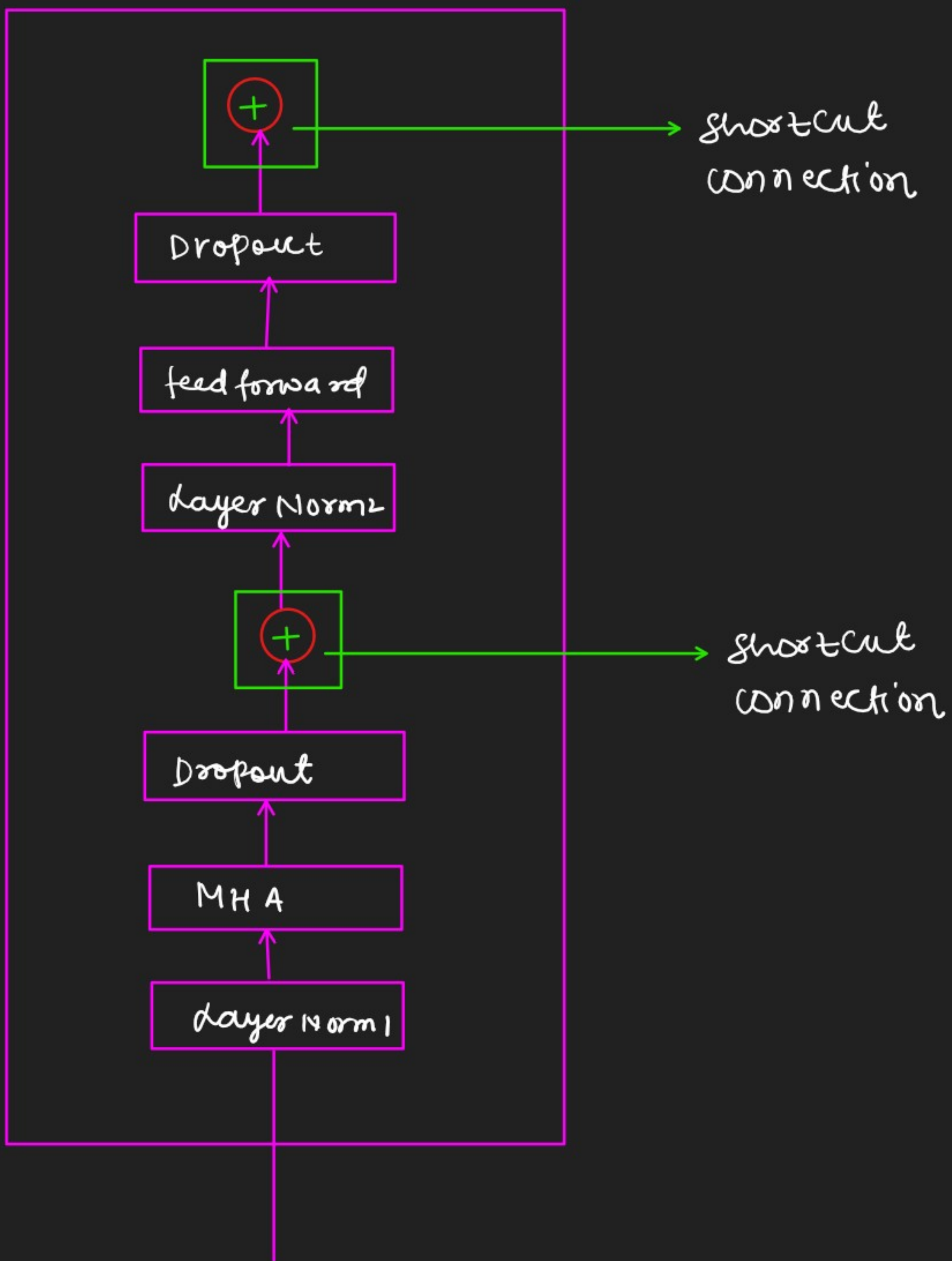


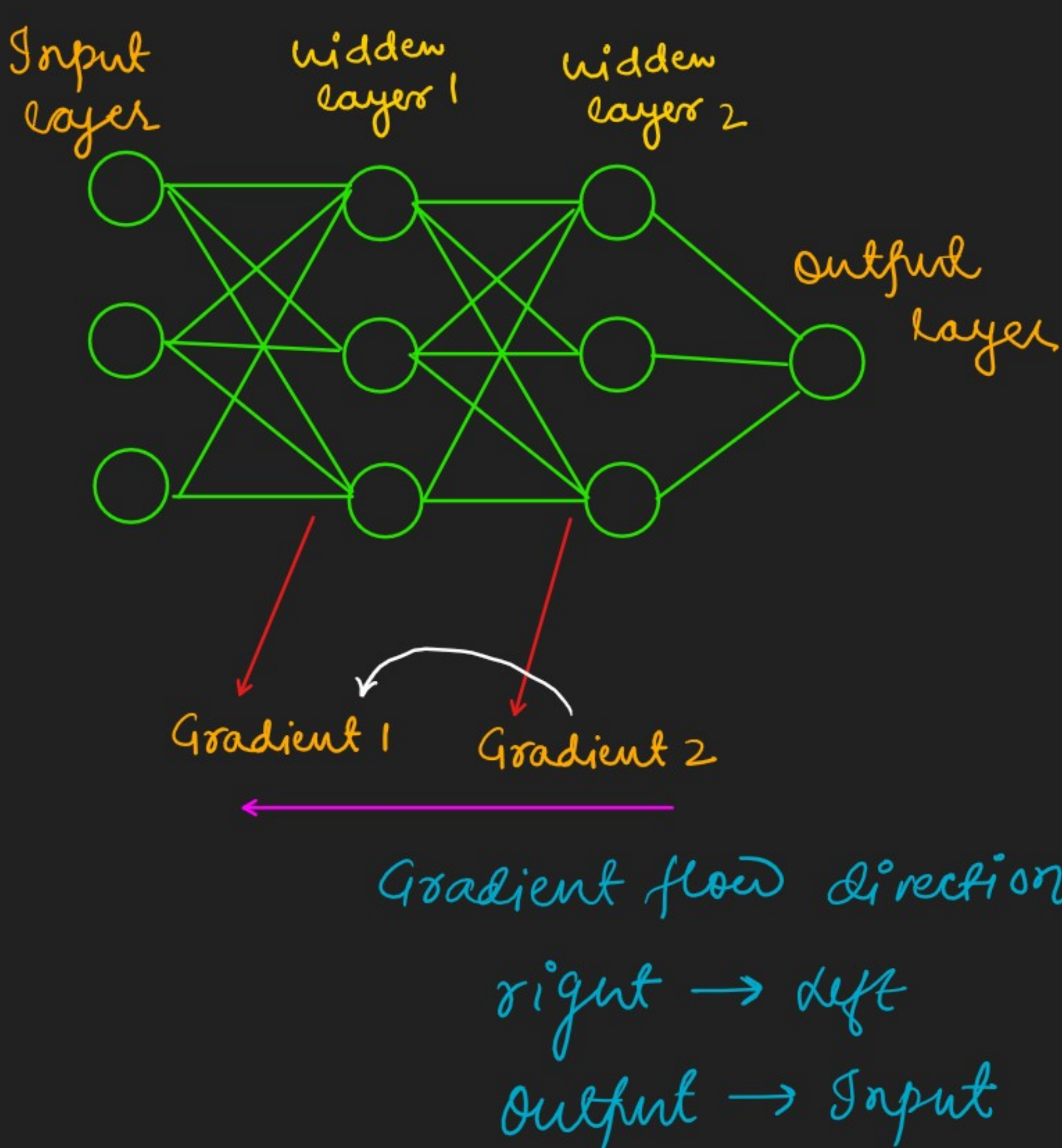
lecture-22: Shortcut connection

Also known as skip or residual connections



- ① shortcut connections were first proposed in the field of computer vision to solve the problem of vanishing gradient.

Gradient becomes progressively smaller as they propagate backward $\rightarrow$ making it difficult to train earlier layers



$$w = w - \alpha \cdot \frac{\partial L}{\partial w} \rightarrow \approx 0$$

i.e. $w = w_{old}$

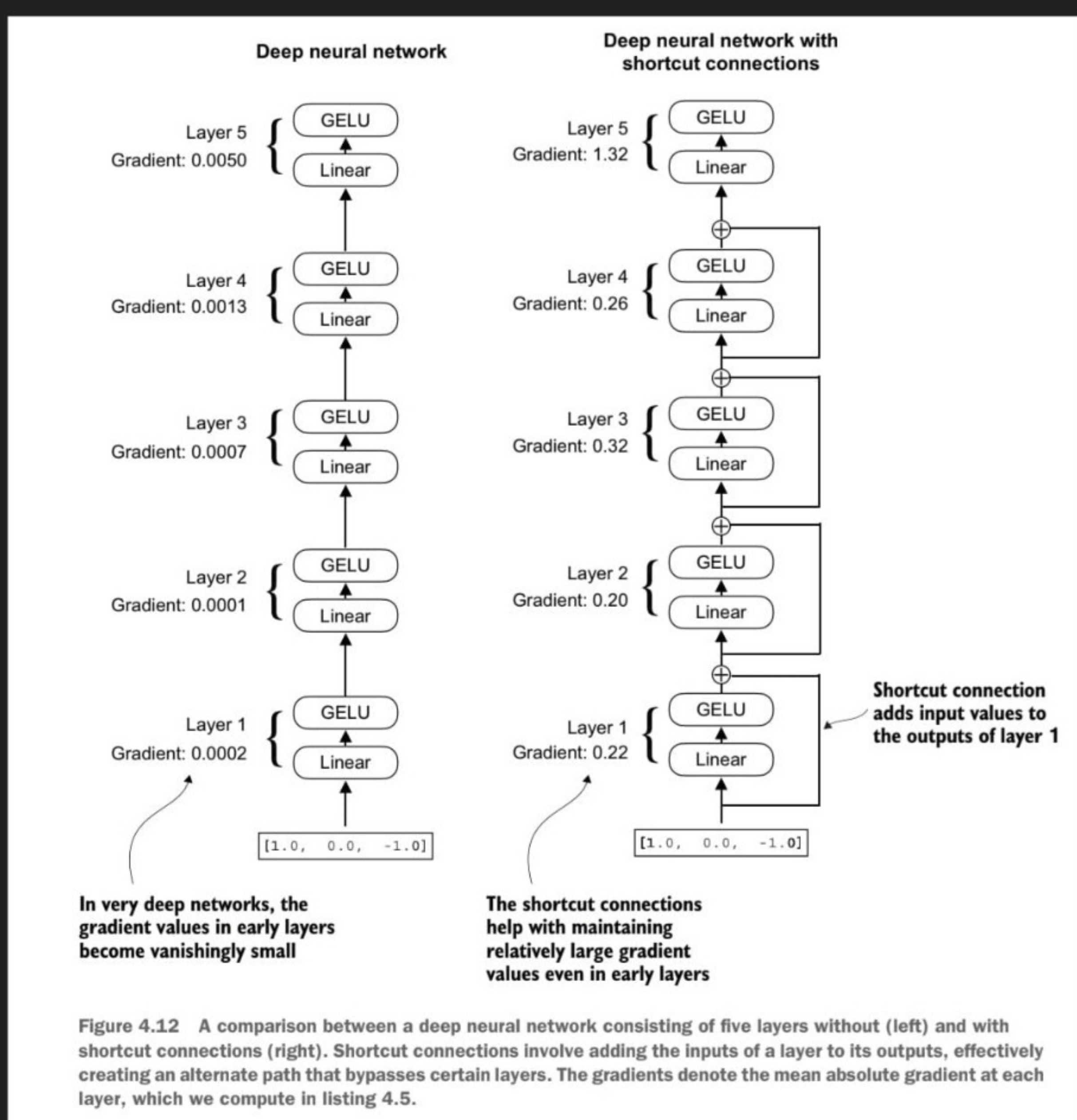
Here weights are not updating
so neural network is not learning

② shortcut connection create an alternative path for the gradient to flow, by skipping one or more layers.

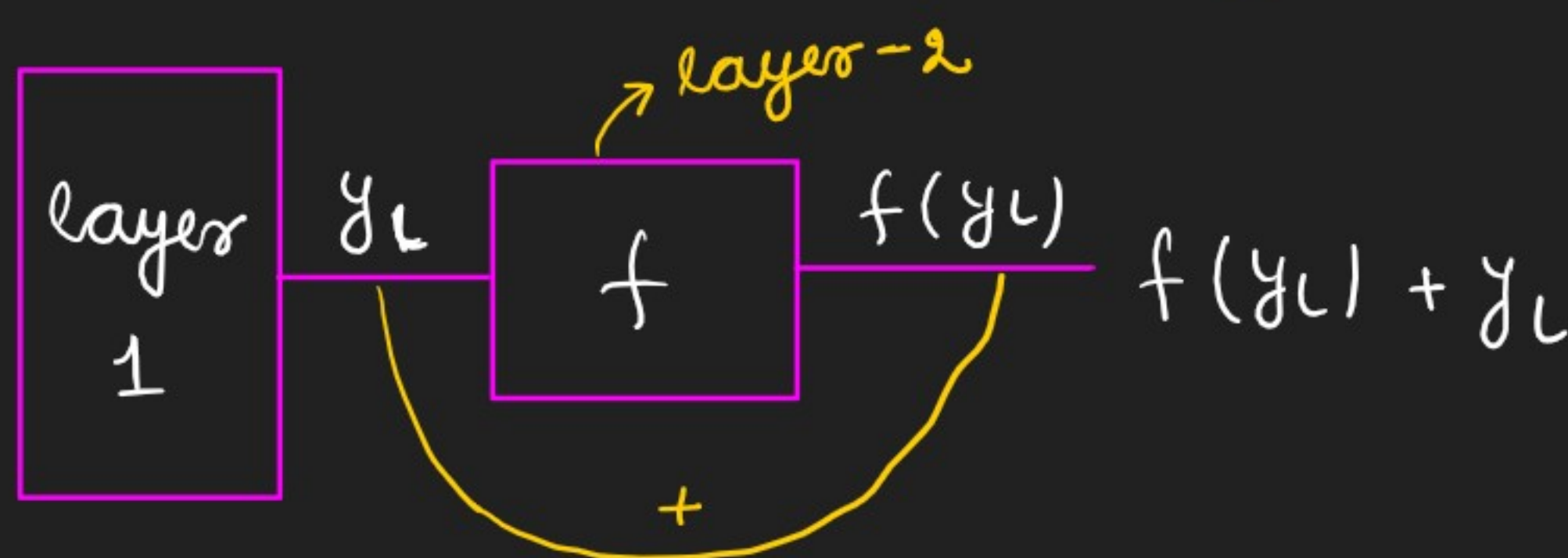
This is achieved by adding the output of one layer to the output of a later layer.

→ This is also called skip connections.

→ play a crucial role in preserving flow of gradient during training backward pass.



Mathematical understanding



$$y_{L+1} = f(y_L) + y_L$$

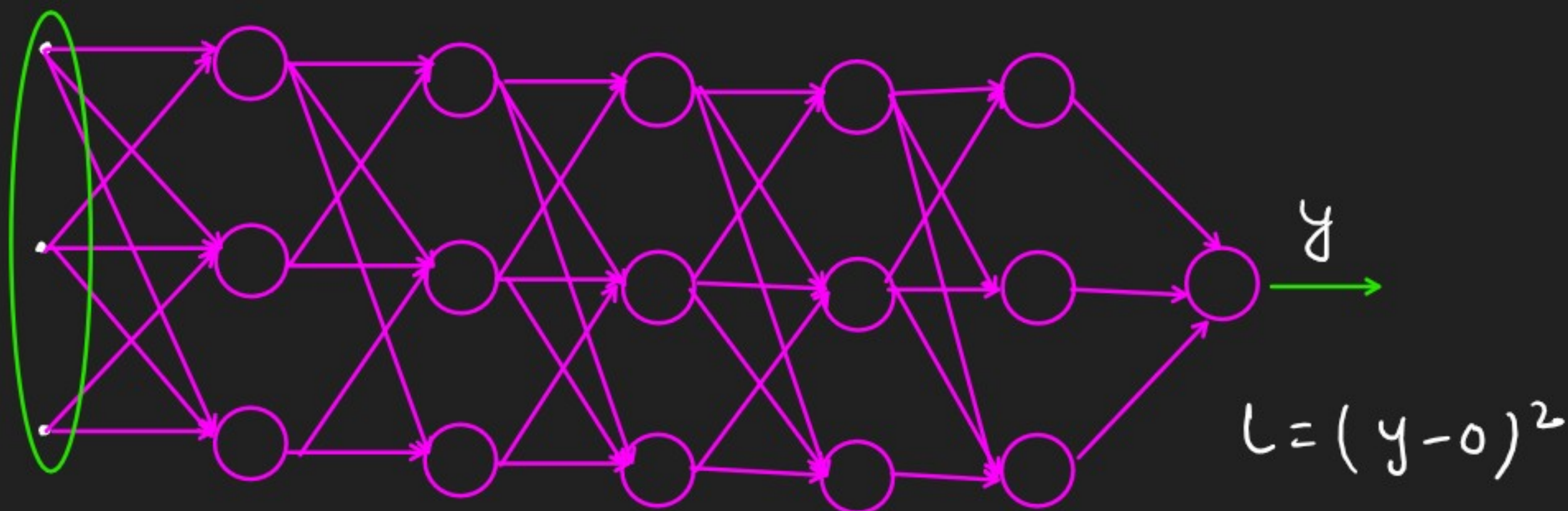
$$\frac{\partial \mathcal{L}}{\partial y_L} = \frac{\partial \mathcal{L}}{\partial y_{L+1}} \left(\frac{\partial y_{L+1}}{\partial y_L} \right)$$

$$\frac{\partial \mathcal{L}}{\partial w}$$

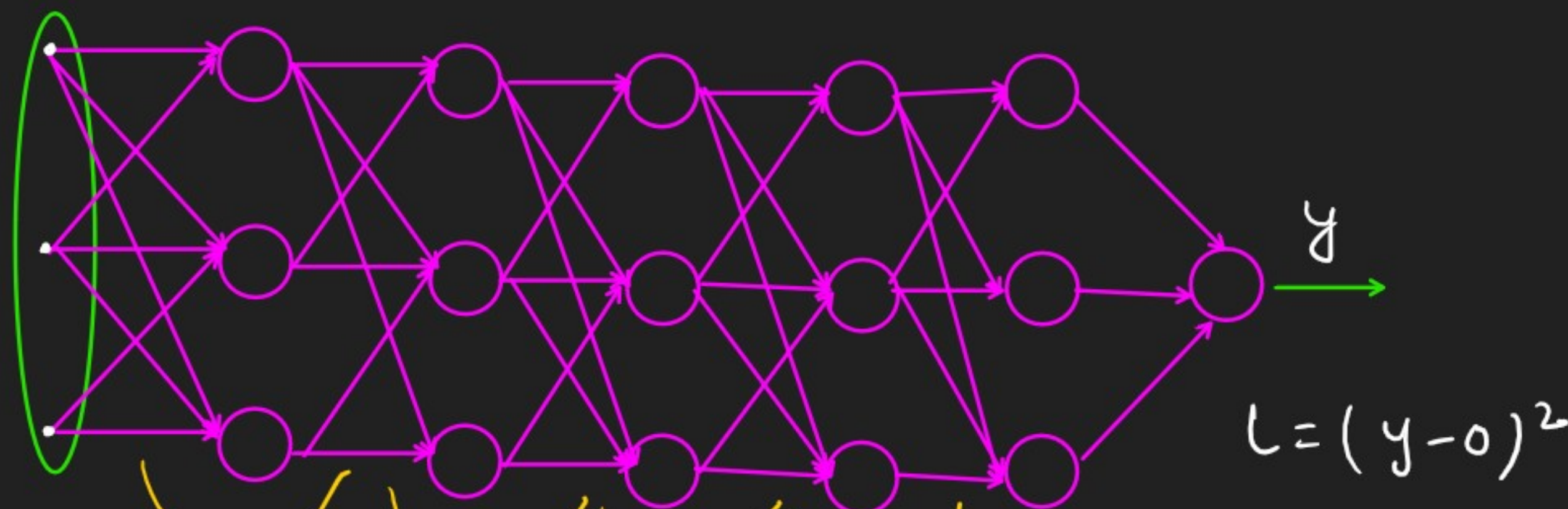
$$\frac{\partial \mathcal{L}}{\partial y_{L+1}} \left(\frac{\partial f(y_L)}{\partial y_L} + 1 \right)$$

≈ 0

keep the gradient through the network.



Input



Input

↑
shortcut
connection